

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B1. FLUIDS

MICHAELMAS TERM 2020

Thursday, 8 October, 09.30 am – 11.30 am

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

Page 2 contains physics formulae and data for this paper.

The questions start on page 3.

The Navier–Stokes equation for viscous, incompressible, fluid flow under gravity is

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} + \frac{1}{\rho} \nabla p + g \mathbf{k} = \nu \nabla^2 \mathbf{u} ,$$

where $\mathbf{u}$ is the fluid velocity, ρ the density, p the pressure, g the acceleration due to gravity, $\mathbf{k}$ the vertical unit vector and ν the kinematic viscosity.

The *Jacobian* $\mathbf{J}$ of a dynamical system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ is defined as $J_{ij} = \frac{\partial f_i}{\partial x_j}$, where f_i and x_j are the i^{th} and j^{th} components of $\mathbf{f}(\mathbf{x})$ and $\mathbf{x}$ respectively. The trace τ of the Jacobian is equal to the sum of its eigenvalues, while the determinant Δ is equal to their product.

For problems in atmospheric physics, unless otherwise stated, x and y refer to the horizontal eastward and northward directions and z the upward direction with corresponding velocities u , v and w .

1000 mb = 1000 hPa = 10^5 Pa.

1. (a) If the gravitational acceleration is negligible, write down the Navier–Stokes equation and mass conservation equation for flow of an incompressible, viscous liquid and provide a physical interpretation for each term. [4]

(b) Consider two-dimensional flow of a viscous fluid driven by oscillations of an infinite horizontal plate, such that $\mathbf{u} = (U \cos \omega t, 0)$ at $z = 0$, where x is the horizontal coordinate and z is the vertical coordinate. The velocity decays to zero far above the plate. The fluid density and viscosity are constant. Explain why we expect to find exact solutions of the Navier Stokes equations of the form $\mathbf{u} = [u_x(z, t), u_z(z, t)]$ and $p = p(z, t)$. By considering mass conservation, determine the solution for u_z . Hence show that u_x satisfies

$$\frac{\partial u_x}{\partial t} = \nu \frac{\partial^2 u_x}{\partial z^2},$$

for kinematic viscosity ν . By considering solutions of the form $u_x = \text{Re}[u_0(z) \exp(i\omega t)]$ or otherwise, determine $u_x(z, t)$ at late enough times for the solution to be approximately independent of the initial condition. [9]

(c) Calculate the four components of the stress tensor, σ_{xx} , σ_{xz} , σ_{zx} and σ_{zz} at time t and depth z in the interior of the fluid domain, to within an additive constant. [4]

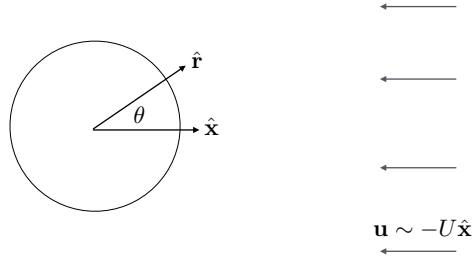
(d) Sketch the variation of u_x with z at times $t = (2n\pi)/\omega$ and $t = (2n + \frac{1}{2})\pi/\omega$ where n is a large integer. Explain the dominant force balances that give rise to the temporal evolution of u_x . Determine an expression for the boundary layer thickness in terms of ν , U and ω . By considering the imposed timescale of the forcing, provide a physical interpretation for this length scale.

Discuss whether this solution could be applied to model oscillatory tidal flow over a fixed stationary boundary. [8]

2. (a) Consider an incompressible and irrotational flow of inviscid fluid with constant density ρ . Explain why the flow can be characterised using a velocity potential ϕ , which satisfies $\nabla^2 \phi = 0$.

Starting from the Euler equations derive the unsteady form of Bernoulli's theorem which shows that $\nabla H = 0$ for an incompressible, inviscid and irrotational flow with constant density and gravitational acceleration g , where the quantity H should be defined. Provide a physical interpretation of the quantity H if, in addition to the above conditions, the flow is also steady. [7]

(b) Consider the corresponding two-dimensional flow past an active cylinder, where internal pumps create a radial velocity component $u_r = A \sin \theta$ on the surface of the cylinder $r = a$, where θ is defined in the figure. The cylinder axis is parallel to the gravitational force. A uniform velocity $\mathbf{u} \rightarrow -U\hat{\mathbf{x}}$ is prescribed in the far-field, where the pressure approaches a constant value $p \rightarrow p_0$.



State the boundary conditions on the velocity potential ϕ , and determine the resulting solution for $\phi(r, \theta)$ and $\mathbf{u}(r, \theta)$. Show that the pressure on the surface of the cylinder is

$$p(a, \theta) = p_0 + \frac{1}{2} \rho U^2 \left[2 \cos 2\theta - 1 - \frac{A^2}{U^2} + 2 \frac{A}{U} \sin 2\theta \right].$$

Sketch $p(a, \theta)$ in the two cases $A = 0$ and $A = U$. [12]

(c) For the analogous problem with a high Reynolds number flow with non-zero viscosity, discuss where boundary layers are likely to form when $A = 0$. By assuming that the pressure variation is similar to the inviscid solution, identify the likely locations of boundary layer separation when $A = 0$. By assuming that the tangential flow near to the cylinder surface is in the same direction as in the inviscid solution, discuss whether the active cylinder with $A = U$ promotes or inhibits boundary layer separation on the upper surface ($\theta > 0$) compared to an inert cylinder with $A = 0$. [6]

[The general azimuthally-periodic solution to $\nabla^2 \phi = 0$ in plane polar co-ordinates is

$$\phi(r, \theta) = A_0 \log r + B_0 + \sum_{n=1}^{\infty} \left(A_n r^n + \frac{B_n}{r^n} \right) \cos n\theta + \left(C_n r^n + \frac{D_n}{r^n} \right) \sin n\theta,$$

for constants A_n, B_n, C_n and D_n].

3. (a) Starting from the Navier–Stokes equation for an incompressible flow with constant density ρ and kinematic viscosity ν , show that the vorticity ω evolves according to

$$\frac{D\omega}{Dt} = (\omega \cdot \nabla)\mathbf{u} + \nu \nabla^2 \omega.$$

Provide a physical interpretation for the terms in this equation.

For inviscid flow in two spatial dimensions with constant density ρ , explain why this reduces to

$$\frac{D}{Dt} \nabla^2 \psi = 0, \quad (1)$$

where ψ is the streamfunction. [8]

(b) Consider a parallel flow comprising of 2 regions with different shear given by

$$u_x(x, y) = \bar{U}(y) = \begin{cases} -\alpha y, & y \geq 0, \\ 0, & y < 0, \end{cases}$$

$$u_y(x, y) = 0,$$

where $y = 0$ represents an interface between the layers of different initial shear. Determine the streamfunction $\psi = \bar{\psi}(y)$, and show that it satisfies equation (1). What is the vorticity in each layer? [4]

(c) Now consider small amplitude disturbances to this background state where the layer interface $y = 0$ is displaced to $y = \eta = \text{Re} [\eta_0 e^{ikx - i\omega t}]$ and the streamfunction and pressure perturbation satisfy

$$\psi = \bar{\psi}(y) + \text{Re} [\hat{\psi}(y) e^{ikx - i\omega t}], \quad p = \text{Re} [\hat{p}(y) e^{ikx - i\omega t}].$$

By linearising the vorticity equation, determine a linear differential equation for $\hat{\psi}(y)$. Using the horizontal momentum equation, determine $\hat{p}$ in terms of ω , k , $\hat{\psi}$, $\bar{U}(y)$, ρ and their derivatives.

Show that the linearised form of the matching conditions at $y = \eta$ yields

$$\left[\hat{\psi} \right]_-^+ = 0, \quad \left[\frac{\omega}{k} \frac{\partial \hat{\psi}}{\partial y} + \hat{\psi} \frac{\partial \bar{U}}{\partial y} \right]_-^+ = 0, \quad \text{at } y = 0.$$

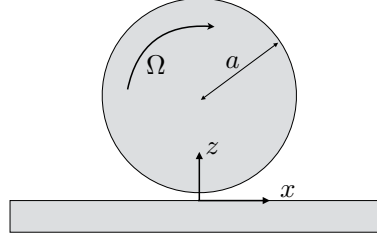
By assuming that $\omega - k\bar{U} \neq 0$ in the region of interest, solve for $\hat{\psi}$ to within a multiplicative constant, and show that

$$\omega = -\frac{\alpha}{2}.$$

Describe how the disturbances evolve in space and over time. [13]

4. (a) Define the *Reynolds number* for a viscous fluid flow. Explain how the physics differs between high and low Reynolds number flows. [4]

(b) A long rigid circular cylinder of length L and radius a lies above a rigid planar surface at $z = 0$, as illustrated in the figure. The cylinder axis is fixed and lies parallel to the planar surface, but the cylinder rotates with angular velocity $\Omega \hat{\mathbf{y}}$. The cylinder and plane are immersed in an incompressible viscous fluid.



(i) Show that near the point of closest separation of the cylinder and plane, the gap height h satisfies

$$h(x) \approx h_0 + \frac{x^2}{2a}, \quad \text{for } x \ll a. \quad (2)$$

where h_0 is gap height at closest separation. State and justify boundary conditions on the fluid velocity $\mathbf{u}$ on the rigid plane and on the surface of the cylinder. Hence explain why $\mathbf{u} \approx U \hat{\mathbf{x}}$ at $z = h(x)$ for $x \ll a$, and determine U .

(ii) Explain why the flow in the gap will be approximately independent of y , and justify that the horizontal velocity component u_x satisfies

$$\mu \frac{\partial^2 u_x}{\partial z^2} \approx -\frac{\partial p}{\partial x},$$

for sufficiently small gap widths, specifying the required conditions on h . By assuming that $\partial p / \partial x$ can be neglected, determine approximate solutions for the horizontal velocity component $u_x(x, z)$. [10]

(c) If the net force $F \hat{\mathbf{x}}$ and torque $\Gamma \hat{\mathbf{y}}$ acting on the cylinder are dominated by viscous stresses in a region $-b < x < b$ near to $x = 0$, show that

$$F \approx \int_{-b}^b \frac{A}{h} dx$$

where A should be specified. Determine a corresponding expression for Γ . By assuming that the integrals can be well approximated using equation (2) for $h(x)$ even for large x , and letting $b \rightarrow \infty$, determine F and Γ explicitly. [5]

(d) Explain how the expressions for F and Γ will change if the cylinder is instead free to roll and slide, rotating with angular velocity $\Omega(t) \hat{\mathbf{y}}$ and translating with velocity $V(t) \hat{\mathbf{x}}$, with both varying slowly over time. If the cylinder has mass M and moment of inertia I , determine approximate equations of motion for the cylinder. Deduce that at large times the cylinder undergoes a pure rolling motion, with $V = \Omega a$. [6]